Galactic Astronomy 2025–2026

Mercè Romero-Gómez



Friedrich Anders



Teresa Antoja (coordinator)



Gaia UB

ICCUB Institute of Cosmos Sciences

Excelencia María de Maeztu

Universitat de Barcelona

How do galaxies form and evolve?

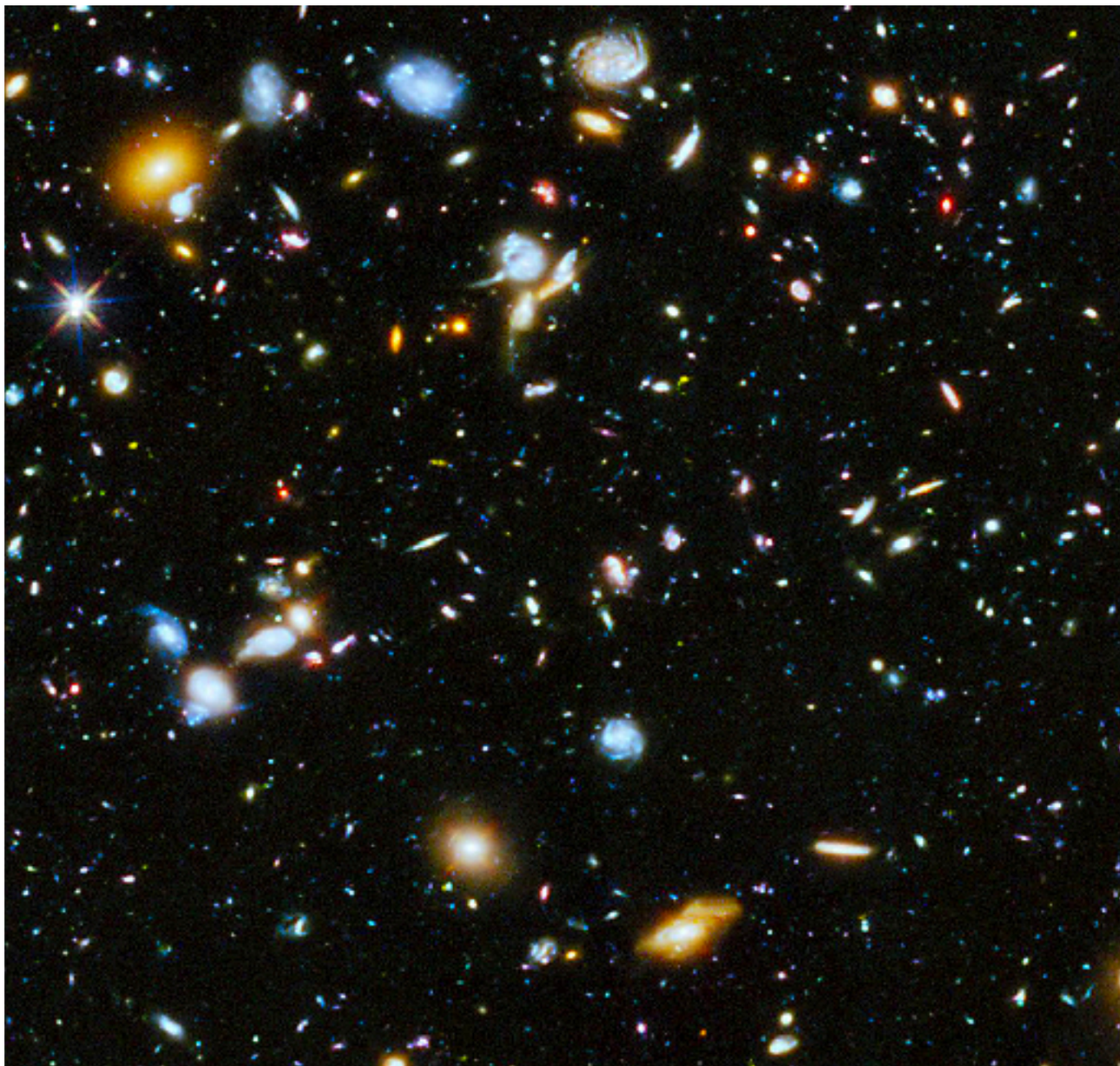
One of the key questions in modern astrophysics:

- A Science Vision for European Astronomy, ASTRONET, 2007
- Science Vision and Infrastructure Roadmap, ASTRONET, 2022

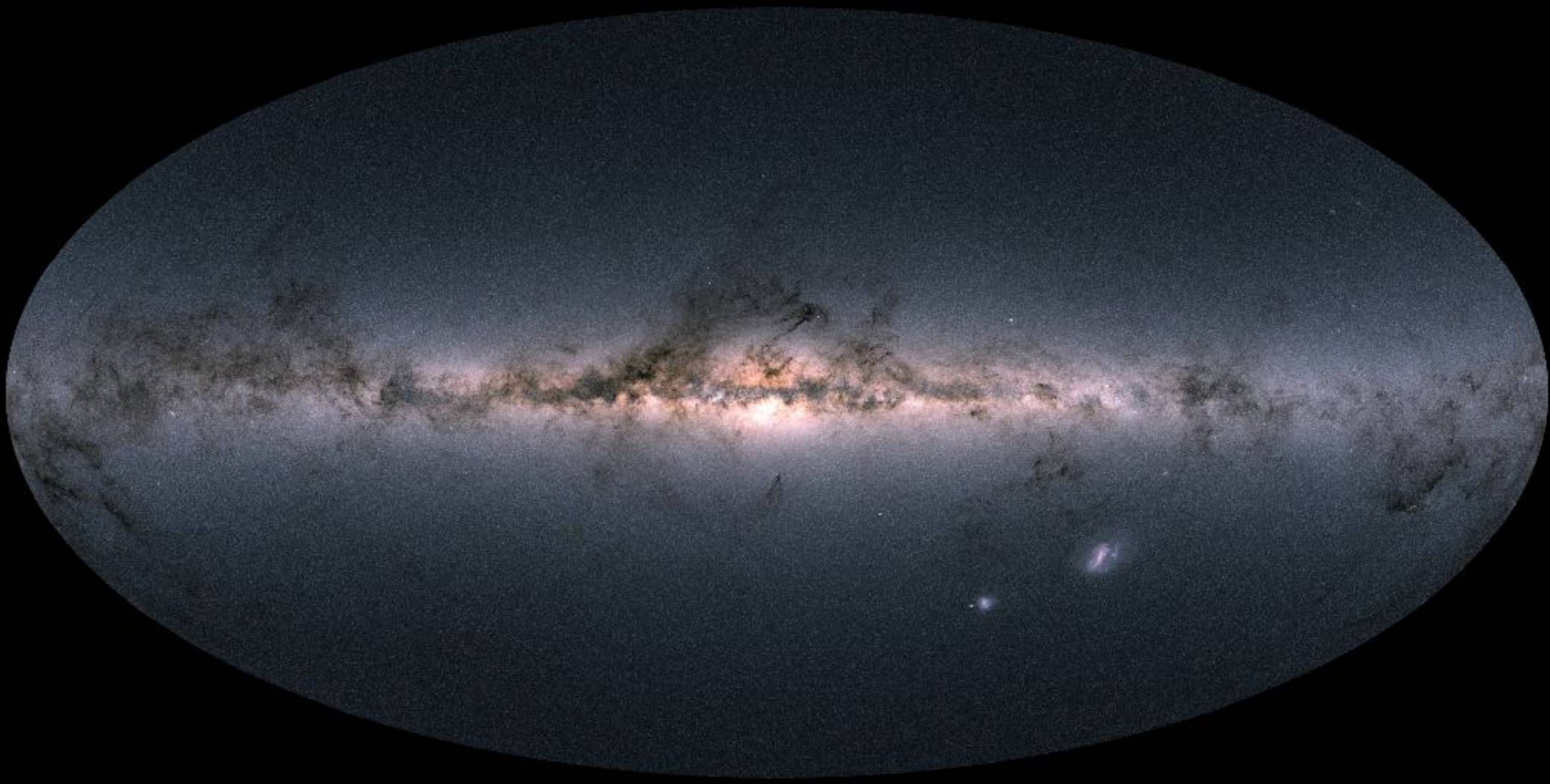
Webb's First Deep Field (NIRCam)



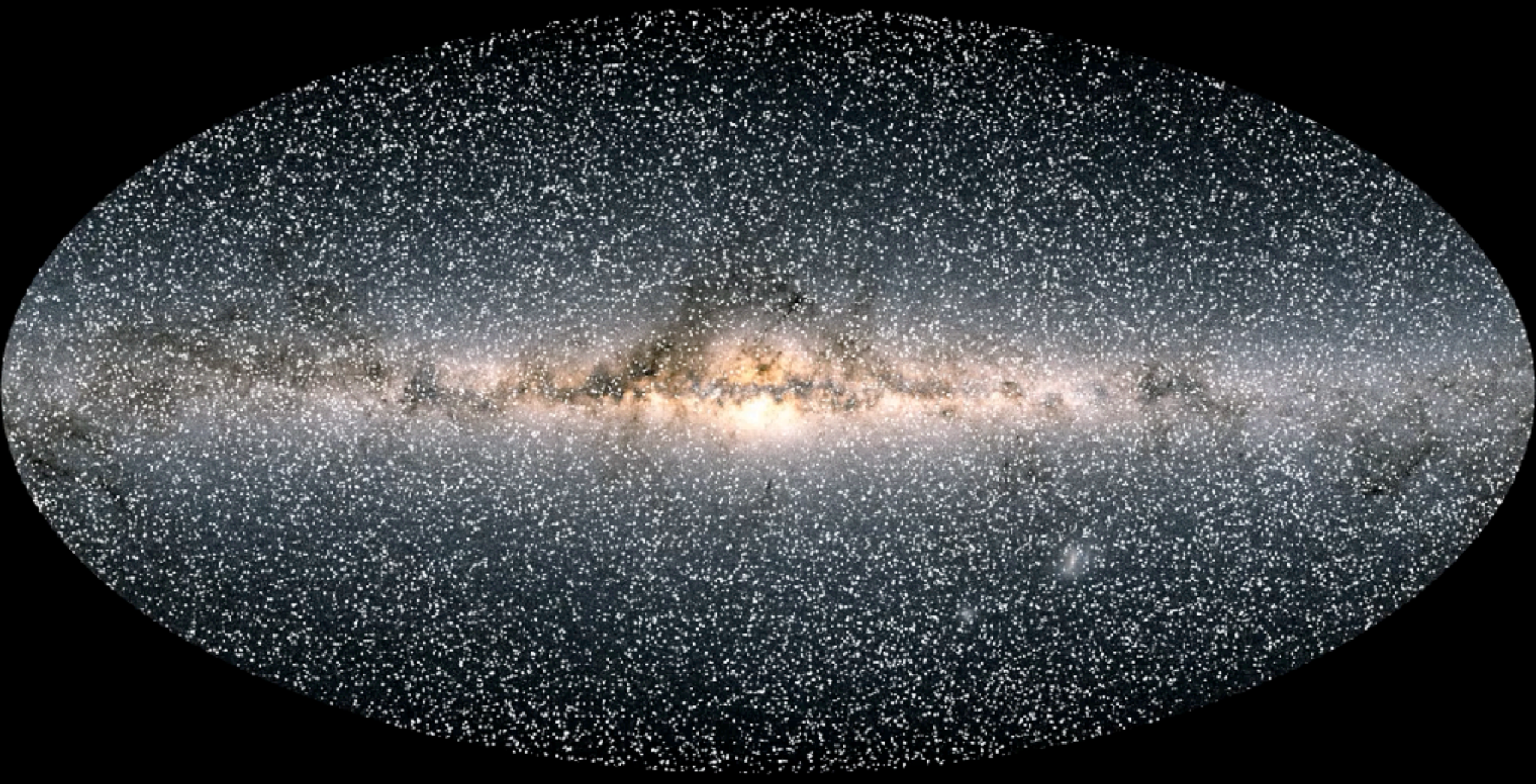
Hubble Ultra Deep Field



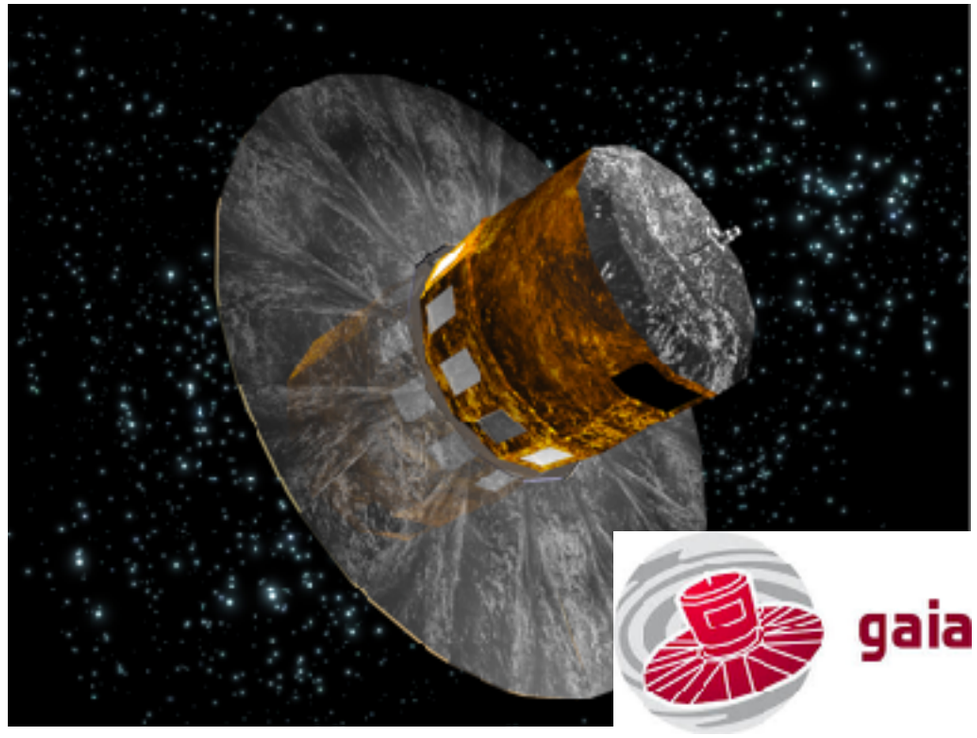
Gaia DR2



Gaia EDR3







December 19th, 2013
10:12 CET

General Objectives

1. Provide a comprehensive and modern overview of the Milky Way, its evolution, and its role in the Universe, informed by Gaia and other large surveys

2. Understand how astronomical observables are used to characterise stars and Galactic structures, and learn their limitations

3. Develop a solid understanding of statistical approaches to stellar populations and Galactic structure

4. Build a foundation in Galactic kinematics and dynamics (collisionless and collisional)

5. Understand the chemical evolution of the Milky Way

What we will see in the course

1. **Introduction TERESA**

2. **Astronomical measurements FEDE**
   1. Astrometry
   2. Photometry
   3. Spectroscopy

3. **Statistical astronomy FEDE**
   1. Apparent distribution of stars
   2. Stellar statistics fundamental equation
   3. Stellar luminosity function
   4. Initial Mass Function and Star formation History
   5. Galactic models for star count predictions
   6. **Exercises observations & measurements**

What we will see in the course

Kinematics & Dynamics

4. **Galactic kinematics** **TERESA**
   1. Galactocentric reference systems
   2. Rotation curve and Oort constants
   3. **Activities rotation**
   4. Kinematics of solar neighbourhood stars
5. **Galactic Dynamics I: basic concepts** **TERESA & MERCE**
   1. Gravitational potentials & Poisson equation
   2. Orbits
   3. ***Hands-on orbits,***
   4. Collisionless dynamics
6. **Galactic dynamics II: advanced** **TERESA & MERCE**
   1. Dynamics of spiral structure and bars
   2. Dynamical modelling
   3. Galaxy interactions, non-axisymmetry, Gaia
   4. ***hands-on kinematics/Gaia recent data***
   5. ***Journal Club***
7. **Collisions and encounters of stellar systems** **SARA**

What we will see in the course

8. **Chemical evolution of the Milky Way FEDE**
   1. Observational evidence
   2. Surface gas density, supernova explosion rate and metal enrichment
   3. Basic elements of a model of chemical evolution
   4. Some simple models

PROGRAM

1. Introduction TERESA
   15 set
   - 1.1. Galaxies and their role in the Universe
   - 1.2. Galactic astronomy history
   - 1.3. Global description of the Milky Way: our present knowledge
2. Astronomical measurements FEDE
   , 16 17 18 set
   - 2.1. Astrometry
   - 2.2. Photometry
   - 2.3. Spectroscopy
3. *+2h Exercises measurements* FEDE *22 23 set*
4. Statistical astronomy FEDE
   25 29 30 set 1 2 oct
   - 4.1. Apparent distribution of stars
   - 4.2. Stellar statistics fundamental equation
   - 4.3. Stellar luminosity function
   - 4.4. Initial Mass Function and Star formation History
   - 4.5. Galactic models for star count predictions
5. Galactic kinematics (Marcin) TERESA
6 7 8 9 oct
   - 5.1. Galactocentric reference systems
   - 5.2. Rotation curve and Oort constants
   - 5.3. Activities rotation
   - 5.4. Kinematics of solar neighbourhood stars
6. Galactic Dynamics I: basic concepts
13 14 15 16 20 21 22 23 27 oct
   - *6.1.* Gravitational potentials & Poisson equation TERESA
   - *6.2.* Orbits TERESA
   - *+2h hands-on orbits* MERCE, *28 29 oct*
   - 6.3. Collisionless dynamics MERCE *30 oct 3 4 5 6 10 nov*
7. Galactic dynamics II: advanced
   - *7.1.* Dynamics of spiral structure and bars *11 12 13 17 18 nov* MERCE
   - 7.2. Dynamical modelling *19 nov* TERESA
   - 7.3. Galaxy interactions, non-axisymmetry, Gaia TERESA *20 24*

+2h hands-on kinematics/Gaia recent data TERESA *25 26 nov*

+1h Journal Club TERESA *27 nov*

   - *7.4.* Collisions and encounters of stellar systems SARA *1 2 3 dec*
8. Chemical evolution of the Milky Way FEDE
   4 9 10 11 15 16 17 18 dec
   - 8.1. Observational evidence
   - 8.2. Surface gas density, supernova explosion rate and metal enrichment
   - 8.3. Basic elements of a model of chemical evolution
   - 8.4. Some simple models

Not 100% definitive

TERESA ANTOJA MERCE ROMERO-GÓMEZ
FRIEDRICH ANDERS SARA RASTRELLO

Activities

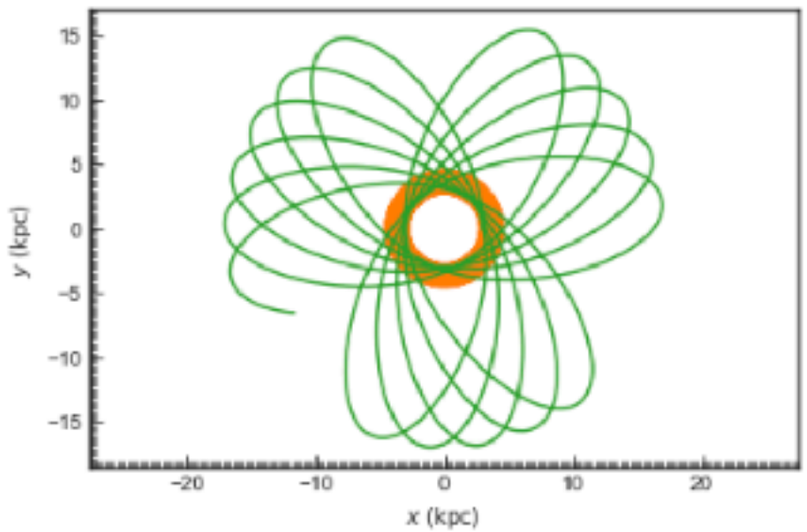
Hands-on exercises

- Gaia Archive & Gaia measurements / Statistical astronomy
- Rotation curves (NEW)
- Orbits in Galactic potentials (deliverable)
- Analysis of *Gaia* kinematic data

In [25]:

```
omw1.plot(d1='x',d2='y')
omw2.plot(d1='x',d2='y',overplot=True)
omw3.plot(d1='x',d2='y',overplot=True)

plt.axis('equal')
plt.show()
```



Paper reading and discussion (journal club)

Lectures from invited researchers



Sara Rastrello



Collisional dynamics

- Basic concepts for collisional systems: relaxation, core collapse, etc
- Dynamics of Globular Clusters
- Nov-Dec

Evaluation

During course

Short tasks + presentations

Hands-on work

Participation

40%

Exam

5/10 required

60%

Exam date: TBD

Campus

https://campusvirtual.ub.edu/

- *Detailed calendar*
- *Pdfs of lectures*
- *Material for hands-on, exercises, etc*
- *Additional material, papers, etc*

Master's thesis in Galactic astronomy

F. Anders, T. Antoja, T. Asano, L. Balaguer, J. M. Carrasco, F. Figueras, M. Gieles, X. Luri, E. Masana, M. Romero-Gomez, M. Semcsuk

- Open clusters
- Globular clusters
- Galactic Disk dynamics
- Satellite galaxies
- Stellar evolution
- Population synthesis
- Big Data and Machine learning
- Galaxy formation and cosmology
- Chemical evolution of the MW
- Spectroscopic stellar surveys
- Light pollution
- White dwarfs





